UK Travel Behavior Analysis & Product Strategy Report

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I. Introduction

This report analyzes the travel behavior of UK residents based on the Travelpac 2022-2023 dataset provided by the UK Office for National Statistics (ONS). The purpose is to support tourism businesses in designing marketing strategies with appropriate products. The dataset includes demographic variables, trip purpose, travel type, means of transport, length of stay, and total expenditure. Through descriptive analysis, confidence interval estimation, hypothesis testing, ANOVA, and multiple regression, I will study and identify the main factors affecting travel spending and tour selection behavior. The results show that age, gender, and tour type are factors that have a significant impact on spending, in which the 55+ group chooses package tours with the highest spending. On that basis, the report proposes marketing strategies and designs new tour products aimed at this customer segment. The entire analysis is presented in a practical application-oriented manner, aiming to provide data evidence to support business decisions and brand positioning in the international tourism industry.

II. Part A - Exploratory Data Analysis

2.1 Descriptive statistic and data patterns

The descriptive statistics summary table shows the mean value of "Visits" falling at 4536.15. While the median of "Visits" is just about 2017.35, a considerable difference. Thus, it can be seen that the "Visit" index tends to have a very right-skewed distribution, up to 10.72. The same is true for the other two continuous variables. "Nights" has a mean of 47754.98 and a median of 17501.83. "Expend" has a mean of 3774690.18 and a median of 1343865.77. Both have very high skewness of 6.88 and 9.27, respectively. It can be seen that all three continuous variables of the Travelpac dataset have a very large positive skewness due to the existence of many trips by a small group of VIP tourists with large and long-term spending but scattered and spread out at the top of the data. As a result, the average of the indicators has been directly raised higher than the majority of the sample. This characteristic is generally quite suitable and coincides with the tourism industry data recorded in previous datasets and the Pareto model commonly seen in consumer data (assuming that 20% of consumers will create 80% of the value). If viewed from a statistical perspective, the large difference between the mean and median shows that the mean will not be a good representative index in finding the central number of the data. Therefore, this dataset will focus on the median and IQR to reduce the influence of outliers. Next, the Interquartile Range of the variables all have a large dispersion, showing that the customer behavior in tourism of UK residents is not consistent between the "Standard" and "VIP" groups. From the economist's perspective, the existence of these outliers is not a data error but a characteristic of the tourism industry with a small group of high-value tourists (ONS, 2023). High-end marketing strategies will be deployed with this group, they are less price sensitive but account for a large proportion of revenue.

2.2 Travel trends and behavior overview

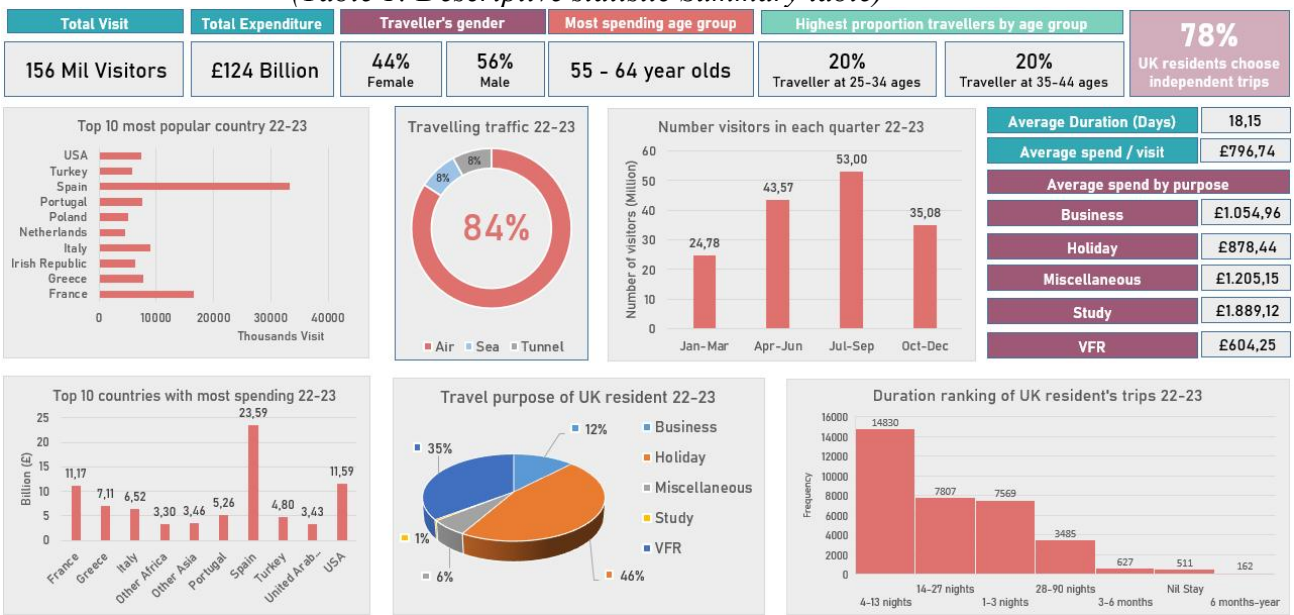
According to the 2022-2023 dashboard, the UK has 156 million outbound visitors with a total spend of £124 billion. The top 5 visiting countries are in Europe but the expenditure of USA visitors is top 2, we can see that UK tourists prefer to go to European countries because of cheaper costs. The majority of visitors choose to travel by air with 84%, only 16% of visitors choose to travel by water and high-speed train. The peak tourist season is recorded in the third quarter (July-September) with 53 million visitors, clearly reflecting the British summer cycle, so marketing strategies should focus on the first and second quarters to attract customers first. In terms of trip length, short breaks lasting 4 to 13 days are the trendiest choice, accounting for about 14,830 trips. And the average trip

length is 18.15 days but the standard deviation is still high (SD=101674.10) showing that long trips still account for a significant proportion. The average spending per trip is £796.74, of which holiday trips account for 46% but only spend an average of £878.44 per visit. In contrast, study trips with £1,889.12 and miscellaneous trips with £1,205.15 have double the spending level compared to the average. This shows that the cost of studying and staying abroad is quite expensive. The business trip group reached £1,054.9/visit, showing that they are willing to spend more to receive convenience and time saving. Finally, the VFR group accounts for 35% but spends the least, showing a trend of saving costs. In addition, in terms of demographics, the 55-64 age group is the highest spending group with a higher proportion of men, suitable as a target for high-end tour packages. In conclusion, the right-skewed distribution in tourism spending reflects the potential for developing premium product packages targeting VIP customers with spending levels far above the average.

Descriptive statistic summary

	Mean	Median	Min	Max	Skewness	SD	Q1	Q3
Visit	4536,151383	2017,355	59,617	289133,58	10,72528883	11286,64014	1043,282	4070,61675
Nights	47754,98302	17501,838	0	2007374,458	6,888658017	101674,107	6808,517	45629,40625
Expend	3774690,181	1343865,771	1060,203	269901174,1	9,27550235	9216809,803	521243,236	3464528,126

(Table 1: Descriptive statistic Summary table)



(Figure 1: EDA Dataset dashboard)

III. Part B - Interval Estimation

To obtain reliable estimates of travel behavior, 95% confidence intervals were constructed for "Average Expenditure per Visit" and "Average Nights per Visit". The dataset has a very skewed-right distribution and contains many outliers, so the median will be computed using a non-parametric approach instead of the mean. This provides a more robust estimate of the central tendency, less sensitive to extreme high spenders. Creating two new columns, "Expend per Visit = Expend/Visit" and "Nights per Visit = Nights/Visit", helps to convert these variables to the same unit per trip, thereby better reflecting the actual travel behavior of customers instead of the raw aggregate as in the dashboard which reflects sum of all dataset. With a large sample size of n=34248, the formula for estimating confidence intervals according to the normal distribution is applied:

Confident Interval				
95% Confident interval → α = 0,05 → z α/2 = 1,96				
	n	Mean (Median)	SD	SE
Expend per visit	34248	627,586	1416,520	7,654
Nights per visit	34248	7,316	24,768	0,134
CI Formula	$\bar{x} \pm Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$			
Expend per visit CI	Upper CI	642,588		
	Lower CI	612,584		
Nights per visit CI	Upper CI	7,578		
	Lower CI	7,054		

(Table 2: Confidence interval estimation)

The results show that the average trip will be between £612,584 and £642,588. The average number of nights per year will be between 7,054 and 7,578. Compared to the overview dashboard above, this figure is significantly lower due to the elimination of the VIP group with high spending influence, which gives a more robust estimate and a more accurate representation of the "Standard" traveler, reducing the bias due to skewness. However, because the two variables "Expend" and "Nights" are still skewed due to many outliers, 6.8 and 9.27, respectively. I decided to use the Bootstrap method (Bias-Corrected and Accelerated - BCa) applied in SPSS with 1,000 simulated samples. This method is used to estimate 95% confidence intervals instead of the standard formula and reduce the impact of outliers without assuming a normal data distribution. The results show that Expend per visit has a 95% CI falling between £985.7 - £1,016.9, and Nights per visit has a 95% CI falling between 14.5 - 15.1 nights. Compared to the dashboard and median robust, this result is lower because the Bootstrap reflects the behavior of "Standard" travelers instead of being pulled up by the high spenders. The median values of £627.6 and 7.3 nights also confirm the right skewness of the data.

From a business perspective, the analysis shows that the majority of UK visitors spend around £600-£1,000 for a 7-15 night trip, while only a small number of high-end visitors drive the average higher. Therefore, tour operators should focus on the middle-spending segment when designing tour packages, and develop premium products specifically for groups with superior spending behavior.

Bootstrap Specifications

Sampling Method	Simple
Number of Samples	1000
Confidence Interval Level	95,0%
Confidence Interval Type	Bias-corrected and accelerated (BCa)

Descriptives

					Bootstrap[a]			
							BCa 95% Confidence Interval	
			Statistic	Std. Error	Bias	Std. Error	Lower	Upper
Expendpervisit	Mean		1001,71844	7,654408	-,00162	7,81281	985,66474	1016,98645
	95% Confidence Interval for Mean	Lower Bound	986,71555					
		Upper Bound	1016,72133					
	5% Trimmed Mean		813,06151		,16290	4,61022	803,92087	822,58822
	Median		627,58600		,41594	4,39397	618,67644	637,99957
	Variance		2006588,866		-6046,649	146015,808	1755824,865	2288075,074
	Std. Deviation		1416,541163		-3,069116	51,396790	1325,115042	1512,507608
	Minimum		1,000					
	Maximum		49999,994					
	Range		49998,994					
	Interquartile Range		837,817		1,435	9,203	818,000	860,207
	Skewness		8,871	,013	-,196	1,282	6,700	10,706
	Kurtosis		182,148	,026	-8,960	49,832	98,666	254,567
Nightspervisit	Mean		14,76629	,133837	,00362	,13748	14,50707	15,05653
	95% Confidence Interval for Mean	Lower Bound	14,50397					
		Upper Bound	15,02862					
	5% Trimmed Mean		10,86924		,00183	,07324	10,72808	11,02163
	Median		7,31600		-,00065	,05855	7,19932	7,41437
	Variance		613,458		,412	24,139	568,188	662,678
	Std. Deviation		24,768080		,003528	,486885	23,836905	25,741110
	Minimum		,000					
	Maximum		365,000					
	Range		365,000					
	Interquartile Range		11,604		,004	,189	11,246	11,962
	Skewness		5,838	,013	-,008	,131	5,578	6,061
	Kurtosis		47,837	,026	-,142	2,331	43,442	51,819

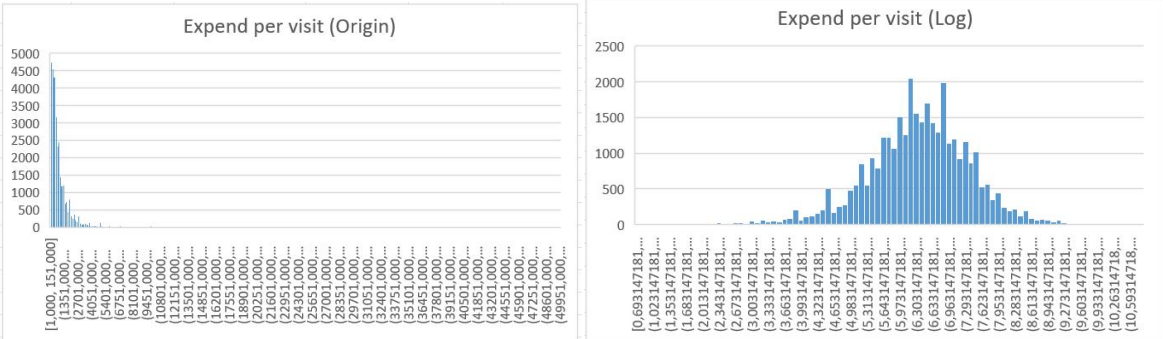
a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

(Table 3: Boostrap result table - SPSS)

IV. Part C - Hypothesis Testing

4.1 Purpose of using logarithmic transformation

The Expenditure per visit variable will be log-transformed to correct the right-skewed distribution due to the existence of many small expenditure values and some very large outliers. Log transformation helps reduce the influence of outliers, making the data closer to normal distribution and stabilizing the variance between groups, thereby meeting important assumptions of ANOVA and linear regression. In addition, the log-transform model allows the interpretation of results in terms of percentage change in expenditure, making the analysis and presentation of economic significance more intuitive and accurate.



(Figure 2: Histograms show the different of log-transformation)

4.2 Hypothesis 1: Differentiation in average expenditure for a trips between males and females

Setting up the hypothesis, we have:

H0: $\mu_1 = \mu_2$ represent for "Mean expenditure is the same for both males and females"

H1: $\mu_1 \neq \mu_2$ represent for "Mean expenditure differs between genders"

"Expend per visit" is a strongly right-skewed variable, so the log-transformation technique will be applied to reduce the influence of outliers.

Hypothesis testing: Expenditure by Genders					
Using Logarith Transformation method for "Spend of Male" and "Spend of Female"					
Hypothesis construct:					
H0: $\mu_1 = \mu_2$	Mean expenditure is the same for both males and females				
H1: $\mu_1 \neq \mu_2$	Mean expenditure differs between genders				
Indicators:					
	n	Mean (Log)	SD	Var	Geo Mean
Spend by male	19096	6,453	1,048	1,099	734,144
Spend by female	14820	6,330	1,045	1,092	647,040
t-statistic method					
t Stat	10,7638				
p value (two-tail)	5.67×10^{-27}				
Mean Difference (log)	0,123				
Back-log transform	1,13				
Different in spend	(1,13-1)x100 = 13,1%				

(Table 4: t-test of hypothesis 1)

The test applied is a two-sample t-test assuming unequal variances. This is an appropriate test when comparing the average of two independent groups with different variability. The test results show that there is a difference in statistical significance between the two genders with t = 10.76 and p-value <0.001, leading to H0 being rejected. More specifically, the average expenditure of males is 13% higher than that of females, there is a clear difference (after back transform). This shows that men tend to spend more on overseas travel and suggests the potential for unique tourism product packages that are oriented to marketing to men.

4.3 Hypothesis 2: Proportion of choosing tour travel differs between age groups

H0 : No association between age group and package preference

H1 : Package preference differs by age group

Age * package Crosstabulation						
			package		Total	Comments
			Independent	Non-Independent		
Age	0-15	Count	976	266	1242	Mostly independent travel, few tours
		% within Age	78,6%	21,4%	100,0%	
	16-24	Count	2793	750	3543	Mostly independent travel, few tours
		% within Age	78,8%	21,2%	100,0%	
	25-34	Count	4939	1299	6238	The highest in independent travel
		% within Age	79,2%	20,8%	100,0%	
	35-44	Count	4983	1362	6345	Mostly independent travel, few tours
		% within Age	78,5%	21,5%	100,0%	
	45-54	Count	4327	1335	5662	Start increasing tour rates
		% within Age	76,4%	23,6%	100,0%	
	55-64	Count	3612	1235	4847	Prefer package
		% within Age	74,5%	25,5%	100,0%	
	65 & over	Count	2618	1101	3719	Clearly biased towards package trips
		% within Age	70,4%	29,6%	100,0%	
Total		Count	26540	7708	34248	At all ages, independent is still more c
		% within Age	77,5%	22,5%	100,0%	

Chi-square test			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi Square	275,586[a]	7	0,000
Likelihood ratio	283,026	7	0,000
N of Valid cases	34248		

a. 0 cells (0,0%) have expected count less than 5. The minimum expected count is 279,53.

(Table 5: Chi-square testing of hypothesis 2)

Both "Age" and "Package" are categorical variables, so the Chi-square test will be applied with SPSS to evaluate the relationship between age and travel type of British people. The results show that the relationship between these two variables is statistically significant with $\chi^2 = 275.59$ and df = 7, thus rejecting H0. More specifically, 79% of the young group aged 16-34 will tend to choose independent travel. In contrast, the older group over 55 will tend to choose package tours. This finding has shown the difference in travel behavior of generations, young people will prioritize personal experiences and exploration while the elderly prefer safety and pre-planned routes.

In conclusion, both tests use methods that produce highly statistically significant results showing statistical significance between variables. Gender will influence tourist spending and age will influence travel patterns. These differences will help us to differentiate customer segmentation and design appropriate products.

V. Part D - ANOVA Analysis

Anova: Single Factor (Log-transform)					
SUMMARY					
Groups	*Count*	*Sum*	*Average*	*Variance*	*Geometric mean*
0-15	1241	7314,297	5,894	1,076	*361,5861816*
16-24	3540	21651,238	6,116	1,205	*451,8382011*
25-34	6234	39738,655	6,375	1,072	*585,3064179*
35-44	6342	40490,545	6,385	1,046	*591,2005466*
45-54	5658	36477,740	6,447	1,112	*629,4539077*
55-64	4839	31582,041	6,527	1,089	*681,5848948*
65 & Over	3717	24341,399	6,549	1,154	*696,8393717*

ANOVA								
Expend (Log-transform)								
			Sum of Squares	df	Mean Square	F	*F crit*	*P-value*
Between Groups	(Combined)		774,259	6	129,043	117,194	2,09888	5,833E-147
	Linear Term	Unweighted	682,485	1	682,485	619,816		
		Weighted	617,735	1	617,735	561,012		
		Deviation	156,524	5	31,305	28,430		
Within Groups			34755,371	31564	1,101			
Total			35529,630	31570				

(Table 6: ANOVA testing)

Test of Homogeneity of Variances					
		Levene Statistic	df1	df2	Sig.
LnExpend	Based on Mean	3,730	6	31564	0,001
	Based on Median	3,473	6	31564	0,002
	Based on Median and with adjusted df	3,473	6	31480,823	0,002
	Based on trimmed mean	3,592	6	31564	0,001

(Table 7: Levene Statistic)

To test whether the average spending per trip abroad has any significant differences within the framework of age groups, this section will use the ANOVA (Analysis of Variance) test. Since the variable "Expend per visit" is a right-skewed variable, I will also continue to use the variable "Expend per visit" that has been logarithmically transformed to flatten the data.

First, the Levene test shows that the variance between groups is different (p-value = 0.001 < 0.05). Therefore, Welch ANOVA, a form of ANOVA, can be used when the variance between groups is not equal. The result from the Anova analysis is $F(6, 31564) = 117.19$ and p-value < 0.001, showing a statistically significant difference between age groups, meaning that age does have an impact on the average spending level.

Posthoc Test - Multiple Comparisons

Dependent Variable: Expend (Log-transform)

Games-Howell

Age group		Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval				Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound						Lower Bound	Upper Bound
1	2	-,222320*	0,034747	0,000	-0,32486	-0,11978	4	1	,490663*	0,032124	0,000	0,39584	0,58548
	3	-,480649*	0,032233	0,000	-0,57579	-0,38550		2	,268343*	0,022477	0,000	0,20205	0,33463
	4	-,490663*	0,032124	0,000	-0,58548	-0,39584		3	0,010014	0,018353	0,998	-0,04411	0,06413
	5	-,553257*	0,032612	0,000	-0,64951	-0,45700		5	-,062594*	0,019010	0,017	-0,11865	-0,00654
	6	-,632715*	0,033046	0,000	-0,73025	-0,53518		6	-,142052*	0,019745	0,000	-0,20028	-0,08383
	7	-,654827*	0,034315	0,000	-0,75609	-0,55356		7	-,164164*	0,021804	0,000	-0,22847	-0,09986
2	1	,222320*	0,034747	0,000	0,11978	0,32486	5	1	,553257*	0,032612	0,000	0,45700	0,64951
	3	-,258329*	0,022634	0,000	-0,32508	-0,19158		2	,330937*	0,023170	0,000	0,26261	0,39927
	4	-,268343*	0,022477	0,000	-0,33463	-0,20205		3	,072608*	0,019195	0,003	0,01601	0,12921
	5	-,330937*	0,023170	0,000	-0,39927	-0,26261		4	,062594*	0,019010	0,017	0,00654	0,11865
	6	-,410394*	0,023777	0,000	-0,48051	-0,34027		6	-,079458*	0,020530	0,002	-0,14000	-0,01892
	7	-,432506*	0,025512	0,000	-0,50774	-0,35727		7	-,101570*	0,022517	0,000	-0,16797	-0,03517
3	1	,480649*	0,032233	0,000	0,38550	0,57579	6	1	,632715*	0,033046	0,000	0,53518	0,73025
	2	,258329*	0,022634	0,000	0,19158	0,32508		2	,410394*	0,023777	0,000	0,34027	0,48051
	4	-0,010014	0,018353	0,998	-0,06413	0,04411		3	,152066*	0,019923	0,000	0,09331	0,21082
	5	-,072608*	0,019195	0,003	-0,12921	-0,01601		4	,142052*	0,019745	0,000	0,08383	0,20028
	6	-,152066*	0,019923	0,000	-0,21082	-0,09331		5	,079458*	0,020530	0,002	0,01892	0,14000
	7	-,174178*	0,021965	0,000	-0,23896	-0,10940		7	-0,022112	0,023141	0,963	-0,09036	0,04613
							7	1	,654827*	0,034315	0,000	0,55356	0,75609
								2	,432506*	0,025512	0,000	0,35727	0,50774
								3	,174178*	0,021965	0,000	0,10940	0,23896
								4	,164164*	0,021804	0,000	0,09986	0,22847
								5	,101570*	0,022517	0,000	0,03517	0,16797
								6	0,022112	0,023141	0,963	-0,04613	0,09036

*. The mean difference is significant at the 0.05 level.

(Table 8: Game-Howell post-hoc)

Games-Howell post-hoc comparisons were used to compare pairs of groups, which showed that the middle-aged (55-64) and older (over 65) groups had significantly higher average spending than younger people. Back-transforming the data using Log showed that the older group spent 30-40% more than the 16-24 group, which is consistent with the above tests.

As such, businesses can view older customers as VIPs with high spending power, suitable for high-end package tours. The younger group should be targeted with flexible and customizable tours, as well as more affordable prices. The ANOVA results demonstrated the differences in spending between age groups, thereby helping businesses propose appropriate marketing campaigns.

VI. Part E - Multiple Regression

The Regression model is built based on the dependent variable "ln(Expend per visit)" to reduce the impact of outliers and independent variables including destination, travel purpose, demographic factors and travel type. With the formula:

EXPENDi=β0+β1(COUNTRYi)+β2(PURPOSEi)+β3(AGEi)+β4(SEXi)+β5(PACKAGEi)+β6(NIGHTSi)+β7(MODEi)+εi

Regression Statistics (Using ln[Expend])	
Multiple R	0,439
R Square	0,192
Adjusted R Square	0,192
Standard Error	0,953
Observations	31571

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	6	6835,367486	1139,227914	1253,18094	0
Residual	31564	28693,85317	0,909068976		
Total	31570	35529,22065			

	Unstandardized Coefficients	*Standardized Coefficients*	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Back-transform*
(Constant)	5,587		0,031	181,005	0,000	5,845	5,962	
country	0,016	0,336	0,000	66,150	0,000	0,058	0,071	0,016
purpose	-0,120	-0,169	0,004	-31,947	0,000	-0,142	-0,100	-0,113
Age	0,062	0,097	0,003	19,187	0,000	0,015	0,016	0,064
Sex	-0,123	-0,058	0,011	-11,474	0,427	-0,015	0,035	-0,116
package	0,245	0,097	0,013	18,472	0,000	-0,147	-0,133	0,277
nights	1,212E-06	0,120	0,000	23,633	0,000	0,000	0,000	0,000
mode	-5,00E-03	-0,003	0,010	-0,050	0,619			-0,005
Note	Predict the actual change in Y when X increases by 1 unit	Compare the relative influence levels between variables						

(Table 9: Regression statisic of Expend per visit and independent variables)

The regression results show that the model has $R^2 = 0.192$, which means that about 19.2% of the variation in average spending per trip is explained by the independent variables. This level is relatively acceptable in the context of this being a tourism dataset. In addition, the ANOVA test has a very large F result (1253.18) and P-value less than 0.05), showing that this Regression model is statistically significant. In more detail, the coefficient β_Country = 0.016 shows that spending increases by an average of 1.6% when travelers choose another country, showing the influence of the destination on costs. β_Age = 0.062 proves that groups of visitors with older ages, meaning higher income and savings accounts, will tend to spend more. The β_Package = 0.245 is the variable with the strongest impact on tourists' spending behavior. Independent tourists tend to spend 27.7% less per trip than those traveling with a tour. In contrast, β_Purpose = -0.12 and β_Sex = -0.123 show that trips to visit relatives and women spend 12% less than other travel purposes and men. The number of travel days and means of transport have small β, which have almost no effect on spending.

From a business perspective, the results imply that travel companies should focus on middle-aged customers and package tour groups because they are high-spending and stable segments. The limitation of the model is that R^2 is low, indicating that many non-quantitative factors (such as income, motivation, or ticket price) have not been taken into account. However, the current model still provides statistical evidence to help businesses clearly identify the main spending drivers of British tourists.

VII. Part F - Marketing Recommendations (10%)

7.1 Identify target customer segments

Through the data from analysis and reporting, it shows that if the implementation focuses on the group of middle-aged tourists aged 55 and over, it will be the most potential customer segment to increase profits.

For example, the results from ANOVA analysis and Games-Howell post-hoc have proven that the age group 55-64 and 65+ have a higher spending level of 30-40% when traveling compared to the remaining age groups. The regression model also confirms that age and type of travel are the factors that have the strongest impact on spending behavior. In addition, men also tend to spend 13% more than women. Combined with the results of Confidence Interval, the majority of travelers spend £600-£1,000 for 7-15 nights but the 55+ age group is overwhelmed. Therefore, businesses should focus on this group of customers, designing more high-end travel products.

7.2 Strategy 1 - “Platinum Comfort & Care”

The first strategy will focus on developing high-end tours for customers 55+ and position the price at mid-premium/premium. This package tour strategy emphasizes safety and convenience. The product is a 10-14 day tour to famous cultural destinations such as Spain, France, Portugal and Italy, suitable for air travel habits (84%) and average vacations of 4-13 nights. All-inclusive tour packages include door-to-door pick-up, 4-5 star hotels, insurance and 24/7 medical support. In addition, there are typical experiences such as cuisine, heritage, and musicals. Positioned in the mid-premium to premium segment, the branding for this strategy is "Platinum Comfort & Care" to build trust. The strategy aims to increase revenue from the 55+ group by 15%, average booking value (ARPB) by 15%, improve the experience and increase the satisfaction score (NPS) of older customers.

7.3 Strategy 2 - “Early-Bird & Loyalty Flywheel”

The goal of this strategy is to change seasonal consumer behavior, maintain stable revenue by encouraging early bookings and increase loyalty targeting price-sensitive customers. EDA data shows that the peak tourism season falls in Q3 with 53 million visitors in 22-23. Therefore, businesses will implement an Early Bird policy for Q1-Q2, offering experiences or upgrades instead of discounts, helping to maintain profit margins. At the same time, the three-level membership program (Silver-Gold-Platinum) rewards based on trip value, adding benefits such as private check-in, special set menus. In particular, integrating a referral mechanism for large groups of customers helps to effectively expand the customer base. Integrating personalized PR for customers who have searched for the company's tours on the website, fanpage... With the motto "Book early and receive many unimaginable incentives". The strategy aims to increase Q1-Q2 early booking rates by 20%, achieve a 30% return guest rate within 12 months and optimize advertising cost efficiency (CPA).

VIII. Part G - Tour Design & Proposal

Serene Continental Odyssey	
Target customer segmentation	The data analysis results show that the group of tourists aged 55 and over is the most potential segment for businesses. Therefore, the new tour is aimed at customers aged 55 and over with stable income, often willing to spend money to receive high-quality cultural and resort tourism experiences, low risk and slow pace.
Destinations	According to TravelPac's dataset in 2022 - 2023, the top 5 most popular destinations are: + Spain - 33 million visit + France - 16 million visit + Italy - 8 million visit + Greece - 7 million visit + Portugal- 7 million visit According to Travelpac data 2022-2023, Spain, Italy and Portugal are among the top 5 most popular destinations for British travellers, and have seen a significant increase in average spending post-COVID-19. These countries have warm climates, diverse cultures, delicious cuisine and convenient air connections (84% of visitors choose air travel), making them ideal for older travellers.

	Product choice: “Platinum Mediterranean Retreat” - a 12 day itinerary exploring three Western European countries, combining cultural sightseeing with luxury relaxation.
Tour features	The tour will be designed to prioritize less travel but more experiences, reflecting the consumer behavior of the 55+ group. At the same time, it is consistent with the analysis results showing that the group choosing the package tour has the highest spending level: - Time: 12 days 11 nights, balancing between sightseeing and relaxation.Itinerary: Madrid - >Barcelona -> Rome - >Lisbon (3 main points, 1 relaxation stop) - Transportation: Direct flight from UK to Madrid, intra-bloc travel by high-speed train or luxury car, return flight from Lisbon - Accommodation: 4-5 star hotel in the center or coastal area, with medical services and 24/7 support - Activities: Visiting UNESCO heritage sites, experiencing food & wine, art workshops, Flamenco music shows, alternating 2 days of “free choice” for guests to rest or shop - Additional services: Door-to-door pick-up, airport fast-track, full insurance, bilingual tour leader, small group ≤ 20 guests
Price Positioning	The tour is positioned in the mid-premium to premium segment, with an expected price of £2,900-£3,200 per person, including all airfares, accommodation, meals, sightseeing and insurance. This price point is in line with the high spending capacity of older customers (average spend is 27% higher according to regression results) and the “value-for-comfort” strategy to emphasize quality, amenities and all-inclusive experiences rather than competing on low prices The goal is to increase revenue from the 55+ segment by 15% and increase average booking value (ARPB) by 10% in the first year.
Channels & Promotions	**I. Platinum Comfort & Care Campaign** - Channel: Website, fanpage and email marketing targeting customer who had searched for tour - Message: “Travel with peace of mind - Enjoy a complete experience - No worries about unexpected costs.” - Activities: Release video testimonials from real customers, emphasizing the safety and comfort of the journey **II. Early-Bird & Loyalty Flywheel** - Promotion: Book early ≥90 days and receive a special experience or room upgrade - Membership program: Three levels (Silver - Gold - Platinum), accumulate points based on tour value, have special incentives for returning customers - Referral Program: Cooperate with retirement clubs and seniors associations to expand customer base **Expected KPI:** - Increase revenue in the 55+ segment by 15% - Increase early booking rate (Q1-Q2) by 20% - Achieve return rate of ≥30% within 12 months

“Serene Continental Odyssey” is a tour built on quantitative data analysis from Parts A-F, clearly demonstrating the connection between statistics and practical business strategy. The tour targets the 55+ group with high spending, prioritizes safety and comfort, and exploits popular Western European destinations with stable travel demand.
The combination of all-inclusive resort style, cultural experiences and high-class care services not only helps businesses increase revenue and customer loyalty but also strengthens brand image in the luxury travel segment.

VIII. Conclusion

The above essay used many statistical methods including descriptive statistics, confidence interval, hypothesis testing, ANOVA and Regression to analyze Travelpac's 2022-2023 travel data, showing that British travel spending is significantly influenced by age, gender and tour type, in which the 55+ group chooses the highest spending package tour. Based on these quantitative results, businesses should focus on this segment through the product "Serene Continental Odyssey", a luxury vacation itinerary in Spain, Italy and Portugal. The analytical model has provided solid statistical evidence, helping to orient marketing strategies, optimize revenue and strengthen brand image in the luxury travel segment.

IX. Reference

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